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Manifold with inter-opening features for use with a pressure transmitter

Abstract

A manifold, a method of making the manifold, and a system for measuring pressure within a fluid flowline, in which the manifold may comprise a manifold body having a first end and a second end. A first fluid conduit and a second fluid conduit may extend through the manifold body and each have an inlet for coupling to a fluid flowline and an outlet for coupling to a pressure transmitter. First and second isolation valve bores may be in fluid communication with the first and second fluid conduits, respectively. A first equalizing bore may be in fluid communication with at least one of the first and second fluid conduits. The inlet openings may be spaced from each other by a first inter-opening distance, and the outlet openings may be spaced from each other by a second inter-opening distance, the first inter-opening distance being different from the second inter-opening distance.

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(7/108); G01L (9/18); G01L (1/127); G01L (17/005); G01L (5/18); G01L (1/245); G01L (21/32); G01L (1/183); G01L (1/2218); G01L (9/0023); G01L (1/243); G01L (23/145); G01L (5/0047); G01L (5/0076); G01L (9/0088); G01L (1/106); G01L (1/10); G01L (9/001); G01L (1/186); G01L (23/223); G01L (25/00); G01L (5/165); G01L (5/226); G01L (9/0017); G01L (1/044); G01L (3/245); G01L (9/005); G01L (1/08); G01L (21/16); G01L (3/1485); G01L (5/0038); G01L (5/162); G01L (5/225); G01L (7/026); G01L (7/065); G01L (9/0014); G01L (1/04); G01L (1/242); G01L (21/24); G01L (3/10); G01L (5/0004); G01L (5/0052); G01L (7/10); G01L (1/00); G01L (1/103); G01L (1/2275); G01L (1/247); G01L (21/02); G01L (21/26); G01L (23/225); G01L (3/102); G01L (3/105); G01L (5/223); G01L (5/24); G01L (7/028); G01L (9/0011); G01L (5/00); G01L (5/0028); G01L (5/243); G01L (1/083); G01L (1/12); G01L (21/36); G01L (23/04); G01L (23/14); G01L (23/30); G01L (3/103); G01L (5/0033); G01L (5/102); G01L (5/133); G01L (5/1627); G01L (5/166); G01L (7/185); G01L (1/255); G01L (21/08); G01L (5/0057); G01L (5/22); G01L (1/042); G01L (1/122); G01L (1/2225); G01L (1/2243); G01L (1/2256); G01L (1/248); G01L (2009/0067); G01L (2009/0069); G01L (21/06); G01L (23/06); G01L (3/00); G01L (3/06); G01L (3/1478); G01L (3/1492); G01L (3/18); G01L (3/24); G01L (3/242); G01L (5/0061); G01L (5/08); G01L (5/10); G01L (5/101); G01L (5/108); G01L (5/16); G01L (5/161); G01L (5/167); G01L (5/28); G01L (1/046); G01L (2009/0066); G01L (2019/0053); G01L (23/085); G01L (23/20); G01L (5/0071); G01L (5/008); G01L (5/06); G01L (5/171); G01L (7/106)

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
4602657	12/1985	Anderson, Jr.	73/1.61	G01L 19/0015
5277224	12/1993	Hutton	137/884	G01L 19/0015
5709247	12/1997	Hutton	137/884	G01L 19/0015
6009758	12/1999	Petrich	73/756	G01L 19/0015
2001/0039964	12/2000	Gul	137/557	G01L 19/0015

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
2239492	12/2005	CA	F15B 13/0814
2347035	12/2007	CA	F16K 27/003
0866959	12/1999	EP	N/A
WO-9722867	12/1996	WO	F15B 13/0814

OTHER PUBLICATIONS

Parker P3M5S™, P4M5S™, P6M5S™ 5-Valve Soft Seat Manifold brochure, 8 pages (2014). cited by applicant
Parker 5 Valve Soft Seat Manifold P3M5S™, P4M5S™, P6M5S™ brochure, 12 pages (2017). cited by applicant

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Background/Summary

RELATED APPLICATION (1) This application claims the benefit of U.S. provisional patent application 63/320,450, filed Mar. 16, 2022 titled "Manifold for Use with a Pressure Transmitter," the entirety of the disclosure of which is hereby incorporated by this reference.

TECHNICAL FIELD
(1) The technical field generally relates to manifolds for coupling a pressure transmitter to a fluid flowline.
BACKGROUND
(2) Pressure transmitters (also known as "pressure transducers") are mechanical devices that measure the expansive force of a fluid. Pressure transmitters are widely used in a number of applications to measure fluid pressure inside fluid flowlines. For example, in the oil or natural gas industries, pressure transmitters may be used to measure pressure inside pipelines or other pressure sensitive equipment to determine the volumetric flow rate of the fluid therein.
(3) Pressure transmitters are typically coupled to the fluid flowline using a manifold which includes one or more valves used to control the passage of fluid between the fluid flowline and the pressure transmitter. In particular, 5-valve natural gas manifolds incorporating two isolation valves, two equalizing valves and a vent valve are well known in natural gas applications. Conventional 5-valve natural gas manifolds comprise inlet openings spaced from each other a distance of 2½ inches and corresponding outlet openings spaced from each other a similar distance thereby requiring a compatible pressure transducer.
SUMMARY
(4) According to one aspect, there is provided a manifold comprising a manifold body having a first end and a second end, a first fluid conduit and a second fluid conduit extending through the manifold body and each having an inlet for coupling to a fluid flowline and an outlet for coupling to a pressure transmitter, first and second isolation valve bores in fluid communication with the first and second fluid conduits, respectively, and a first equalizing bore in fluid communication with at least one of the first and second fluid conduits, wherein the inlet openings are spaced from each other by a first inter-opening distance, and the outlet openings are spaced from each other by a second inter-opening distance, the first inter-opening distance being different from the second inter-opening distance.
(5) In some implementations, the first inter-opening distance may be greater than the second inter-opening distance and each of the first and second fluid conduits may comprise an inlet-side conduit segment extending form the first end, the inlet-side conduit segments being parallel to each other, and an outlet-side conduit segment that extends toward the second end, the outlet-side conduit segments being angled inwardly toward each other. The first fluid conduit may be a mirror image of the second fluid conduit and the outlet-side conduit segments may be angled towards each other at an angle being between 160° and 170°. Each of the first and second isolation valve bores may

communicate with the inlet-side conduit segment of the respective first fluid conduit and the second fluid conduit. Moreover, the first and second isolation valve bores may extend through and beyond the inlet-side conduit segment of the respective first fluid conduit and the second fluid conduit and into the manifold body. The manifold may further comprise first and second plug bores in fluid communication with the outlet-side conduit segment of the respective first fluid conduit and the second fluid conduit.
(6) In some implementations, each of the first and second fluid conduits may extend along a straight line between the inlet openings and the outlet openings. The manifold may further comprise a second equalizing bore in fluid communication with at least one of the first and second fluid conduits, a vent bore in fluid communication with the first and second equalizing bores, and a vent aperture in fluid communication with the vent bore.
(7) In some implementations, the first isolation valve bore and the first plug bore may extend into the manifold body from a first same face of the manifold body, and the second isolation valve bore and the second plug bore may extend into the manifold body from a second same face of the manifold body that is opposed to the first same face. The first and second isolation valve bores may be located proximate to the first end of the manifold body that is an inlet end, and the first and second plug bores may be located proximate to the second end of the manifold body that is an outlet end.
(8) In accordance with another aspect, there is provided a method of measuring pressure within a fluid flowline, the method comprising utilizing the manifold as described above coupled to the fluid flowline and at least one pressure transmitter, wherein inlet flowlines are respectively coupled to the inlet openings, outlet fluid flowlines are respectively coupled directly to the outlet openings without an adapter between the outlet openings and the outlet fluid flowlines, and the outlet fluid flowlines are in fluid communication with the pressure transmitter.
(9) In accordance with yet another aspect, there is provided a method of manufacturing a manifold, the method comprising providing a manifold body, and boring a first fluid conduit and a second fluid conduit through the manifold body, wherein boring the first and second fluid conduits comprises boring first and second parallel inlet-side conduit segments into a flowline-side body face of the manifold body, and boring first and second angled outlet-side conduit segments into a transmitter-side body face of the manifold body to intersect the first and second parallel inlet-side conduit segments.
(10) In some implementations, the method may further comprise boring first and second isolation valve bores to fluidly communicate with the first and second parallel inlet-side conduit segments, respectively. The method may further comprise boring first and second plug bores to fluidly communicate with the first and second angled outlet-side conduit segments, respectively, and boring the first and second angled outlet-side conduit segments comprises boring the first and second angled outlet-side conduit segments away from each other.
(11) In accordance with another aspect still, there is provided a system for measuring pressure within a fluid flowline, the system comprising the manifold as described above, inlet flowlines respectively coupled to the inlet openings of the manifold for providing fluid thereto, and outlet fluid flowlines respectively coupled directly to the outlet openings of the manifold without an adapter therebetween, the outlet fluid flowlines being in fluid communication with the pressure transmitter for providing fluid thereto.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) FIG. **1** is a top, first side perspective view of a manifold for use with a pressure transmitter, in accordance with one embodiment.

(2) FIG. **2** is a bottom, second side perspective view of the manifold illustrated in FIG. **1**.
(3) FIG. **3** is a bottom plan view of the manifold illustrated in FIG. **1**.
(4) FIG. **4** is a top plan view of the manifold illustrated in FIG. **1**.
(5) FIG. **5** is a first long side elevation view of the manifold illustrated in FIG. **1**.
(6) FIG. **6** is a second long side elevation view of the manifold illustrated in FIG. **1**.
(7) FIG. **7** is a short side elevation view of the manifold illustrated in FIG. **1**.
(8) FIG. **8** is a cross-sectional view, taken along cross-section line A-A, of the manifold illustrated in FIG. **1**.
(9) FIG. **9** is another cross-sectional view, taken along cross-section line B-B, of the manifold illustrated in FIG. **1**.
(10) FIG. **10** is a cross-sectional view, taken along cross-section line C-C, of the manifold illustrated in FIG. **1**.
(11) FIG. **11** is a cross-sectional view, taken along cross-section line D-D, of the manifold illustrated in FIG. **1**.
(12) FIG. **12** is a photograph showing a manifold assembly, in accordance with one embodiment, comprising the manifold illustrated in FIG. **1** and a plurality of valves operatively coupled to the manifold.
(13) FIG. **13** is a long side elevation view of a manifold for use with a pressure transmitter, in accordance with a second embodiment.
(14) FIG. **14** is a cross-sectional view, taken along cross-section line E-E, of the manifold illustrated in FIG. **13**.
(15) FIG. **15** is a cross-sectional view, taken along cross-section line F-F, of the manifold illustrated in FIG. **13**.
(16) FIG. **16** is a cross-sectional view, taken along cross-section line G-G, of the manifold illustrated in FIG. **13**.

DETAILED DESCRIPTION

(17) In the following description, the same numerical references refer to similar elements. Furthermore, for the sake of simplicity and clarity, namely so as to not unduly burden the figures with several references numbers, not all figures contain references to all the components and features, and references to some components and features may be found in only one figure, and components and features of the present disclosure which are illustrated in other figures can be easily inferred therefrom. The embodiments, geometrical configurations, materials mentioned and/or dimensions shown in the figures are optional, and are given for exemplification purposes only.
(18) Furthermore, in the context of the present description, it will be considered that all elongated objects will have an implicit "longitudinal axis" or "centerline", such as the longitudinal axis of a shaft for example, or the centerline of a biasing device such as a coiled spring, for example, and that expressions such as "connected" and "connectable", "secured" and "securable", "engaged" and "engageable", "installed" and "installable" or "mounted" and "mountable", may be interchangeable, in that the present manifold body also relates to kits with corresponding components for assembling a resulting fully-assembled and fully-operational manifold.
(19) It is to be understood that the phraseology and terminology employed herein is not to be construed as limiting and are for descriptive purpose only. The principles and uses of the teachings of the present invention may be better understood with reference to the accompanying description, figures and examples. It is to be understood that the details set forth herein do not construe a limitation to an application of the invention. Furthermore, it is to be understood that the invention can be carried out or practiced in various ways and that the invention can be implemented in embodiments other than the ones outlined in the description above.
(20) Moreover, components of the manifold and/or steps of the method(s) described herein could be modified, simplified, altered, omitted and/or interchanged, without departing from the scope of

the present disclosure, depending on the particular applications which the present manifold is intended for, and the desired end results, as briefly exemplified herein and as also apparent to a person skilled in the art.
(21) Moreover, although the present invention was primarily designed for connecting a fluid flowline to a pressure transmitter, it may be used with other objects and/or in other types of applications, as apparent to a person skilled in the art.
(22) To provide a more concise description, some of the quantitative expressions given herein may be qualified with the term “about”. It is understood that whether the term “about” is used explicitly or not, every quantity given herein is meant to refer to an actual given value, and it is also meant to refer to the approximation to such given value that would reasonably be inferred based on the ordinary skill in the art, including approximations due to the experimental and/or measurement conditions for such given value.
(23) In the following description, the term “about” means within an acceptable error range for the particular value as determined by one of ordinary skill in the art, which will depend in part on how the value is measured or determined, i.e. the limitations of the measurement system. It is commonly accepted that a 10% precision measure is acceptable and encompasses the term “about”.
(24) Referring to FIGS. **1** to **11**, there is shown a manifold **100** for use with a pressure transmitter (not shown). The manifold **100** is usable to couple the pressure transmitter to a fluid flowline (not shown) carrying pressurized fluid, such as a pipeline, and provide fluid communication between the pressure transmitter and the fluid flowline to allow the pressure transmitter to be used to measure fluid pressure or any other relevant parameter of fluid in the fluid flowline.
(25) In the illustrated embodiment, the manifold **100** includes a manifold body **102** which extends between a first end **104** and a second end **106**. In one embodiment, the first end **104** is connectable to the pressure transmitter and the second end **106** is connectable to the fluid flowline.
(26) Specifically, the manifold body **102** includes a transmitter-side body face **108** located at the first end **104** and configured to interface with a corresponding face of the pressure transmitter, and a flowline-side body face **110** located at the second end **106** and configured to interface with a corresponding face of the fluid flowline.
(27) In the illustrated embodiment, the transmitter-side and flowline-side body faces **108**, **110** are planar to extend respectively against the corresponding faces of the pressure transmitter and of the fluid flowline which are also planar. In other embodiments, the transmitter-side and flowline-side body faces **108**, **110** may not be fully planar and could instead include one or more recess, raised portion, concavely or convexly curved portion, or have any other configuration that would be suitable to interface with the corresponding face of the pressure transmitter and of the fluid flowline. In other embodiments still, the transmitter-side and flowline-side body faces **108**, **110** may comprise any shape suitable for fluidly connecting to one or more conduits, the one or more conduits being in fluid communication with one of the pressure transmitter and the fluid flowline.
(28) In the illustrated embodiment, the transmitter-side and flowline-side body faces **108**, **110** extend substantially parallel to each other but alternatively, the transmitter-side and flowline-side body faces **108**, **110** could instead be angled relative to each other.
(29) In the illustrated embodiment, the manifold body **102** includes a transmitter-side flange **200** located at the first end **104**, a flowline-side flange **300** located at the flowline-side end **106** and a central body portion **400** extending between the transmitter-side and flowline-side flanges **200**, **300**. The transmitter-side flange **200** is substantially rectangular and includes first and second long sides **202**, **204** and first and second short sides **206**, **208** that extend parallel to each other and perpendicular to the first and second long sides **202**, **204**. The transmitter-side body face **108** of the manifold body **102** is defined on the transmitter-side flange **200** and extends between the sides **202**, **204**, **206**, **208** of the top flange **200**. Alternatively, the transmitter-side flange **200** could instead have any other shape that would be suitable for connection to the pressure transmitter.
(30) The flowline-side flange **300** is also substantially rectangular and includes first and second

long sides **302**, **304** and first and second short sides **306**, **308** that extend parallel to each other and perpendicular to the first and second long sides **302**, **304**. The flowline-side body face **110** of the manifold body **102** is defined on the flowline-side flange **300** and extends between the sides **302**, **304**, **306**, **308** of the flowline-side flange **300**. Alternatively, the flowline-side flange **300** could instead have any other shape that would be suitable for connection to the fluid flowline.
(31) In the illustrated embodiment, the transmitter-side and flowline-side flanges **200**, **300** are substantially similarly shaped and have substantially the same footprint. Specifically, the transmitter-side flange **200** has a width W.sub.1, defined as a distance between the long sides **202**, **204** of the transmitter-side flange **200**, which is substantially equal to a width W2 of the flowline-side flange **300**, defined as a distance between the long sides **302**, **304** of the flowline-side flange **300**. Similarly, the transmitter-side flange **200** has a length L.sub.1, defined as a distance between the short sides **206**, **208** of the transmitter-side flange **200**, which is substantially equal to a length L.sub.2 of the flowline-side flange **300**, defined as a distance between the short sides **306**, **308** of the flowline-side flange **300**.
(32) In the illustrated embodiment, the transmitter-side flange **200** includes a plurality of connection portions **250** that allow the manifold body **102** to be secured to the pressure transmitter. Specifically, the plurality of connection portions **250** includes four fastener-receiving openings **252** defined in the transmitter-side flange **200**. Each fastener-receiving opening **252** is sized, shaped and positioned to receive a fastener—such as a bolt or the like—which engages the pressure transmitter to secure the pressure transmitter to the manifold **100**.
(33) As best shown in FIGS. **1** to **4**, in the illustrated embodiment, each fastener-receiving opening **252** is shaped as a notch **254** defined in a corresponding long side **202**, **204** of the transmitter-side flange **200**. The notch **254** includes an end portion **256** which is spaced from the corresponding long side **202**, **204** of the transmitter-side flange **200** and a neck portion **258** which extends between the end portion **256** and the corresponding long side **202**, **204**.
(34) It will be appreciated that providing notches **254** instead of closed-side openings that would completely encircle a fastener extending through the opening may allow more flexibility during the manufacturing of the manifold: while closed-side openings would require the openings to be precisely positioned in alignment with corresponding fastener-receiving holes in the pressure transmitter, the notches **254** allow the fasteners to engage the manifold body **102** even if the fasteners are spaced slightly from the end portion **256** of the notches **254**. The notches would also allow the fasteners to be slid laterally into the notches through the neck portion if desired.
(35) The flowline-side flange **300** further includes a plurality of connection portions **350** that allow the manifold body **102** to be secured to the fluid flowline. In the illustrated embodiment, the connection portions **350** include four threaded openings for receiving corresponding threaded fasteners. Alternatively, the connection portions **350** could instead include notches similar to the notches **254**.
(36) Referring now to FIGS. **8** to **11**, the manifold **100** further includes a plurality of fluid conduits **500**, **502** for allowing passage of fluid from the fluid flowline to the pressure transmitter. The fluid conduits **500**, **502** extend through the central body portion **400** and the flanges **200**, **300** between first and second ends **104**, **106** of the manifold body **102**, and more specifically between the transmitter-side and flowline-side body faces **108**, **110**. Specifically, the manifold **100** includes a plurality of inlet openings **520**, **522** defined on the flowline-side body face **110** and a plurality of outlet openings **530**, **532** defined on the transmitter-side body face **108**. In the illustrated embodiment, the inlet openings include first and second inlet openings **520**, **522** and the outlet openings include first and second outlet openings **530**, **532**. Still in the illustrated embodiment, the plurality of conduits **500**, **502** includes a first fluid conduits **500**, **502** extending between the first and second ends of the manifold body **102**.
(37) In the illustrated embodiment, the inlet openings **520**, **522** and the outlet openings **530**, **532** are substantially located in a common transversal plane. Specifically, the center of the inlet openings

520, **522** and the center of the outlet openings **530**, **532** are located in a transversal plane P.sub.1 which extends substantially parallel to the long sides **202**, **204**, **302**, **304** of the transmitter-side and flowline-side flanges **200**, **300** and is located substantially midway between the long sides **202**, **204** of the transmitter-side flange **200** and between the long sides **302**, **304** of the flowline-side flange **300**. The fluid conduits **500**, **502** further extend on this transversal plane P.sub.1.
(38) In the illustrated embodiment, each fluid conduit **500**, **502** has a substantially circular cross-section, and therefore each one of the inlet openings **520**, **522** and the outlet openings **530**, **532** are also substantially circular. Alternatively, each fluid conduit **500**, **502** may have a differently-shaped cross-section and each one of the inlet and outlet openings could have a correspondingly-shaped cross-section.
(39) In the illustrated embodiment, the first and second fluid conduits **500**, **502** further have a constant cross-section along their entire length. Specifically, the first and second fluid conduits **500**, **502** remain substantially circular along their entire length and have a constant diameter—and therefore a constant cross-sectional area—along their entire length. In one embodiment, the first and second fluid conduits **500**, **502** have a diameter of less than about 1 inch, or specifically of between about ¼ inch and ½ inch, or more specifically of about ⅜ inch. Alternatively, the first and second conduits could have any other suitable diameter or vary in diameter along the length of the conduits **500**, **502**.
(40) As best shown in FIG. **2**, in the illustrated embodiment, the inlet openings **520**, **522** and the outlet openings **530**, **532** are spaced apart by different distances. Specifically, the inlet openings **520**, **522** are spaced from each other by a first inter-opening distance D.sub.1 and the outlet openings **530**, **532** are spaced from each other by a second inter-opening distance D.sub.2. It will be understood that the first and second inter-opening distances D.sub.1, D.sub.2 correspond to center-to-center distances between the inlet openings **520**, **522** and the outlet openings **530**, **532**.
(41) In the illustrated embodiment, the first inter-opening distance D.sub.1 is greater than the second inter-opening distance D.sub.2, such that the inlet openings are spaced further apart than the outlet openings. More specifically, in one embodiment, the inlet openings **520**, **522** are spaced apart by a first inter-opening distance D.sub.1 of more than 2 inches, or specifically of between about 2 inches and 2¼ inches, or more specifically of about 2⅛ inches. Alternatively, the inlet openings **520**, **522** may be spaced apart by any other suitable inter-opening distance.
(42) In one embodiment, the outlet openings **530**, **532** are spaced-apart by a second inter-opening distance D.sub.2 of less than 1.5 inches, or specifically of between about 0.5 inch and 1.5 inches, or more specifically of about 1.3 inches. Alternatively, the outlet openings **530**, **532** may be spaced apart by any other suitable inter-opening distance.
(43) It will be appreciated that providing a first inter-opening distance D.sub.1 being different than the second inter-opening distance D.sub.2 may allow the coupling of an otherwise incompatible or unsuitable pressure transmitter to a fluid flowline.
(44) As shown in FIG. **8**, in the illustrated embodiment, the fluid conduits **500**, **502** do not extend along a straight line between the inlet openings **520**, **522** and the outlet openings **530**, **532**. Specifically, each fluid conduit **500**, **502** includes an inlet-side conduit segment **510** located towards the corresponding inlet opening **520**, **522** and an outlet-side conduit segment **512** located towards the corresponding outlet opening **530**, **532**. Each conduit segment **510**, **512** is substantially straight and the inlet-side and outlet side conduit segments **510**, **512** are angled relative to each other to form an elbow **514** between the two segments **510**, **512**.
(45) In the illustrated embodiment, the first conduit segments **510** of the two fluid conduits **500**, **502** extend substantially parallel to each other. More specifically, each first conduit segment **510** has a first segment axis A.sub.1, and the first conduit segments **510** are oriented such that the first segment axes A.sub.1 of the two fluid conduits **500**, **502** extend substantially parallel to each other and substantially orthogonal to the flowline-side body face **110**. Each second conduit segment **512** has a second segment axis A.sub.2 which is angled relative to the first segment axis A.sub.1. It will

be appreciated that the substantially straight positioning of the first conduit segments **510** relative to the flowline-side planar portion **412** may facilitate cleaning or maintenance of said first conduit segments **510**.

(46) Still in the illustrated embodiment, the manifold body **102** is substantially symmetrical about a symmetry plane S extending substantially parallel to the short sides **206**, **208**, **306**, **308** of the transmitter-side and flowline-side flanges **200**, **300** and located substantially midway between the short sides **206**, **208**, **306**, **308**. In this configuration, the second conduit segments **512** of both the first and second fluid conduits **500**, **502** are angled towards the opposite fluid conduit **500**, **502** at the same elbow angle θ such that the two fluid conduits **500**, **502** are mirror images of each other. In one embodiment, the outlet-side conduit segment **512** of the first and second fluid conduits **500**, **502** are angled towards the opposite fluid conduit **500**, **502** at an elbow angle θ of less than 180°, or specifically between about 95° and 175°, or more specifically of about 166.7°. Alternatively, the first and second conduit segments **510**, **512** could be angled relative to each other at any other suitable angle. It will be appreciated that the length of the first conduit segment **510** and the elbow angle θ of the elbow **514** define the difference between the first and second inter-opening distances D.sub.1 and D.sub.2.

(47) In another embodiment, each conduit **500**, **502** may include more than two conduit segments which may be angled relative to each other. In yet another embodiment, one or more of the conduit segments could be curved instead of being straight. In still another embodiment, the two fluid conduits **500**, **502** may not be mirror images of each other and may instead be configured differently.

(48) In the illustrated embodiment, each inlet opening **520**, **522** is further provided with a counterbore **524** extending into the flowline-side body face **110**. Alternatively, the inlet openings **520**, **522** may not be provided with counterbores. In one embodiment, the outlet openings could also include counterbores. Alternatively, the manifold body **102** may not include any counterbores.

(49) In the illustrated embodiment, the central body portion **400** of the manifold body **102** includes first and second lateral sides **402**, **404** extending opposite each other, and third and fourth lateral sides **406**, **408** extending opposite each other.

(50) In the illustrated embodiment, each one of the first and second lateral sides **402**, **404** has a width that is substantially similar to the lengths L.sub.1, L.sub.2 of the transmitter-side and flowline-side flanges **200**, **300**, while the third and fourth lateral sides **406**, **408** have a width which is substantially smaller than the widths W1, W2 of the transmitter-side and flowline-side flanges **200**, **300**.

(51) In the illustrated embodiment, the first lateral side **402** is not entirely planar and instead includes transmitter-side and flowline-side planar portions **410**, **412** that are substantially planar and extend substantially parallel to each other, but that are offset relative to each other to define a shoulder **414** between the transmitter-side and flowline-side planar portions **410**, **412**. The second lateral side **404** is also not entirely planar and includes a planar base portion **416** and a raised portion **418** which extends away from the base portion **416**. In the illustrated embodiment, the raised portion **418** includes first, second and third faces **420**, **422**, **424** which are planar and are angled relative to the base portion **416**, with the third face **424** being located between the first and second faces **420**, **422**. Still in the illustrated embodiment, the third and fourth lateral sides **406**, **408** are substantially planar and extend substantially parallel to each other.

(52) The manifold **100** further includes a plurality of ports or valve bores **600** defined in the manifold body **102** to receive valves for controlling fluid flow within the manifold **100**. For example, FIG. **12** shows a manifold assembly **50** which includes the manifold **100** and a plurality of valves **650** coupled to the manifold body **102**. In the illustrated embodiment, the valves **650** include first and second isolation valves **652**, **654** in communication with the first and second fluid conduits **500**, **502** to selectively open and close the corresponding fluid conduit **500**, **502**. The valves **650** further include first and second equalizing valves **656**, **658** to selectively open and close

corresponding first and second equalizing conduits **560**, **562** allowing communication between the first and second fluid conduits **500**, **502**. Finally, the valves **650** include a vent valve **660** in fluid communication with the first and second equalizing conduits **560**, **562**.

(53) In the illustrated embodiment, the valve bores **600** includes first and second isolation valve bores **602**, **604** defined in the third and fourth lateral sides **406**, **408** of the central body portion **400**, near the flowline-side flange **300**, for receiving the isolation valves **652**, **654**. In a preferred embodiment, the first and second isolation valve bores **602**, **604** intersect the first and second fluid conduits **500**, **502** along the first conduit segments **510**. An orthogonal intersection of the isolation valves **652**, **654** within the first and second fluid conduits **500**, **502** may provide a more effective obstruction of said first and second fluid conduits **500**, **502** when exposed to a high-pressure fluid flow. As best shown in FIG. **9**, the isolation valve bores **602**, **604** are tapered, but could instead be cylindrical.

(54) In the illustrated embodiment, the valve bores **600** further include first and second plug bores **606**, **608** also defined in the third and fourth lateral sides **406**, **408**, near the transmitter-side flange **200**, for receiving first and second plugs **662**, **664**. The first and second plugs **662**, **664** may provide fluid access to the first and second fluid conduits **500**, **502** for the exhausting, collection or sampling of the fluid.

(55) In the illustrated embodiment, the valve bores **600** further include first and second equalizing bores **610**, **612** for receiving the equalizing valves **656**, **658** and a vent bore **614** for receiving the vent valve **660**. Specifically, the first and second equalizing bores **610**, **614** are defined in the first and second faces **420**, **422** of the raised portion **418** and the vent bore **614** is defined in the third face **424** of the raised portion **418**. Referring to FIGS. **5** and **11**, the manifold **100** further comprises a vent port **668** on the flowline-side planar portion **412** opposite the vent bore **614**. The vent port **668** may be in fluid communication with the first and second equalizing conduits **560**, **562** when the vent valve **660** is in an open configuration.

(56) It will be understood that the above description of the bores **600** is merely provided as an example and that bores **600** and valves **650** could be positioned and configured according to any suitable configuration.

(57) It will be appreciated that a closing of the first and second isolation valves **652**, **654** as well as the first and second equalizing valves **656**, **658** may allow a selective isolation of the outlet-side conduit segment **512** for zero-pressure testing of the pressure transmitter.

(58) In the illustrated embodiment, the valve bores **600** are threaded to receive a similarly-threaded portion of the corresponding valve **650**. Alternatively, the valve bores **600** could be unthreaded and instead be configured to be coupled to the valves **650** using other coupling types.

(59) Referring now to FIGS. **13** to **15**, there is there is shown a second embodiment of a manifold **800** for use with a pressure transmitter (not shown). The manifold **800** similarly usable to couple the pressure transmitter to a fluid flowline (not shown) carrying pressurized fluid and provide fluid communication between the pressure transmitter and the fluid flowline.

(60) In the illustrated embodiment, the manifold **800** includes a manifold body **802** which extends between a first end **804** and a second end **806**. In one embodiment, the first end **804** is connectable to the pressure transmitter and the second end **804** is connectable to the fluid flowline.

(61) Specifically, the manifold body **802** includes a transmitter-side body face **808** located at the first end **804** and configured to interface with a corresponding face of the pressure transmitter, and a flowline-side body face **810** located at the second end **806** and configured to interface with a corresponding face of the fluid flowline.

(62) Referring now to FIGS. **14** to **16**, the manifold **800** further includes a plurality of fluid conduits **850**, **852** for allowing passage of fluid from the fluid flowline to the pressure transmitter. The fluid conduits **850**, **852** extend between the first and second ends **804**, **806** of the manifold body **802**, and more specifically between the transmitter-side and flowline-side body faces **808**, **810**. Specifically, the manifold **800** includes a plurality of inlet openings **820**, **822** defined on the

flowline-side body face **810** and a plurality of outlet openings **830**, **832** defined on the transmitter-side body face **808**. In the illustrated embodiment, the inlet openings include first and second inlet openings **820**, **822** and the outlet openings include first and second outlet openings **830**, **832**. Still in the illustrated embodiment, the plurality of conduits **850**, **852** includes a first fluid conduits **850**, **852** extending between the first and second ends of the manifold body **802**, **804**.

(63) As best shown in FIG. **14**, in the illustrated embodiment, the inlet openings **820**, **822** and the outlet openings **830**, **832** are spaced apart by different distances. Specifically, the inlet openings **820**, **822** are spaced from each other by a first inter-opening distance D.sub.1 and the outlet openings **830**, **832** are spaced from each other by a second inter-opening distance D.sub.2. It will be understood that the first and second inter-opening distances D.sub.1, D.sub.2 correspond to center-to-center distances between the inlet openings **820**, **822** and the outlet openings **830**, **832**. In the illustrated embodiment, the first inter-opening distance D.sub.1 is greater than the second inter-opening distance D.sub.2, such that the inlet openings are spaced further apart than the outlet openings. Once again, it will be appreciated that providing a first inter-opening distance D.sub.1 being different than the second inter-opening distance D.sub.2 may allow the coupling of an otherwise incompatible or unsuitable pressure transmitter to a fluid flowline.

(64) As shown in FIG. **14**, in the illustrated embodiment, the fluid conduits **850**, **852** extend along a straight line between the inlet openings **820**, **822** and the outlet openings **830**, **832**. Specifically, each fluid conduit **850**, **852** includes a single conduit segment **855** being substantially straight and extending between the inlet openings **820**, **822** and the outlet openings **830**, **832**. It may be appreciated that the substantially straight conduit segment **855** may facilitate a manufacturing of the manifold **800** by requiring a single bore operation during the formation of the fluid conduits **850**, **852**.

(65) In the illustrated embodiment, the conduit segments **850** of the two fluid conduits **850**, **852** are mirrored and extend at an angle θ.sub.1 and an angle θ.sub.2 relative to the transmitter-side and flowline-side body faces **808**, **810**, respectively. It will be appreciated that, when the transmitter-side and flowline-side body faces **808**, **810** are parallel, the angles θ.sub.1, θ.sub.2 are equivalent.

(66) While the above description provides examples of the embodiments, it will be appreciated that some features and/or functions of the described embodiments are susceptible to modification without departing from the spirit and principles of operation of the described embodiments. Accordingly, what has been described above has been intended to be illustrative and non-limiting and it will be understood by persons skilled in the art that other variants and modifications may be made without departing from the scope of the invention as defined in the claims appended hereto.

Claims

1. A manifold comprising: a manifold body having a first end and a second end; a first fluid conduit and a second fluid conduit extending through the manifold body and each having an inlet for coupling to a fluid flowline and an outlet for coupling to a pressure transmitter; first and second isolation valve bores in fluid communication with the first and second fluid conduits, respectively; and a first equalizing bore in fluid communication with at least one of the first and second fluid conduits; wherein the inlet openings are spaced from each other by a first inter-opening distance, and the outlet openings are spaced from each other by a second inter-opening distance, the first inter-opening distance being different from the second inter-opening distance.
2. The manifold of claim 1, wherein the first inter-opening distance is greater than the second inter-opening distance.
3. The manifold of claim 1, wherein each of the first and second fluid conduits comprises an inlet-side conduit segment extending form the first end and wherein the inlet-side conduit segments are parallel to each other, and an outlet-side conduit segment that extends toward the second end and wherein the outlet-side conduit segments are angled inwardly toward each other.

4. The manifold of claim 3, further comprising first and second plug bores in fluid communication with the outlet-side conduit segment of the respective first fluid conduit and the second fluid conduit.
5. The manifold of claim 3, wherein the first fluid conduit is a mirror image of the second fluid conduit.
6. The manifold of claim 3, wherein the outlet-side conduit segments are angled towards each other at an angle being between 160° and 170°.
7. The manifold of claim 3, wherein each of the first and second isolation valve bores communicates with the inlet-side conduit segment of the respective first fluid conduit and the second fluid conduit.
8. The manifold of claim 7, wherein the first and second isolation valve bores extend through and beyond the inlet-side conduit segment of the respective first fluid conduit and the second fluid conduit and into the manifold body.
9. The manifold of claim 1, wherein each of the first and second fluid conduits extends along a straight line between the inlet openings and the outlet openings.
10. The manifold of claim 1, further comprising a second equalizing bore in fluid communication with at least one of the first and second fluid conduits.
11. The manifold of claim 10, further comprising a vent bore in fluid communication with the first and second equalizing bores.
12. The manifold of claim 11, further comprising a vent aperture in fluid communication with the vent bore.
13. The manifold of claim 1, wherein the first isolation valve bore and a first plug bore extend into the manifold body from a first same face of the manifold body, and the second isolation valve bore and a second plug bore extend into the manifold body from a second same face of the manifold body that is opposed to the first same face.
14. The manifold of claim 13, wherein the first and second isolation valve bores are located proximate to the first end of the manifold body that is an inlet end, and the first and second plug bores are located proximate to the second end of the manifold body that is an outlet end.
15. A method of measuring pressure within a fluid flowline, the method comprising utilizing the manifold as described in claim 1 coupled to the fluid flowline and at least one pressure transmitter; wherein inlet flowlines are respectively coupled to the inlet openings, outlet fluid flowlines are respectively coupled directly to the outlet openings without an adapter between the outlet openings and the outlet fluid flowlines, and the outlet fluid flowlines are in fluid communication with the pressure transmitter.
16. A system for measuring pressure within a fluid flowline, the system comprising: the manifold as described in claim 1; inlet flowlines respectively coupled to the inlet openings of the manifold for providing fluid thereto; and outlet fluid flowlines respectively coupled directly to the outlet openings of the manifold without an adapter therebetween, the outlet fluid flowlines being in fluid communication with the pressure transmitter for providing fluid thereto.
17. A method of manufacturing a manifold, the method comprising: providing a manifold body; and boring a first fluid conduit and a second fluid conduit through the manifold body; wherein boring the first and second fluid conduits comprises: boring first and second parallel inlet-side conduit segments into a flowline-side body face of the manifold body; and boring first and second angled outlet-side conduit segments into a transmitter-side body face of the manifold body to intersect the first and second parallel inlet-side conduit segments.
18. The method of claim 17 further comprising boring first and second isolation valve bores to fluidly communicate with the first and second parallel inlet-side conduit segments, respectively.
19. The method of claim 17 further comprising boring first and second plug bores to fluidly communicate with the first and second angled outlet-side conduit segments, respectively.
20. The method of claim 17, wherein boring the first and second angled outlet-side conduit

segments comprises boring the first and second angled outlet-side conduit segments away from each other.